TGS-2024043854 · DAY 2 · BONUS LAB · 60 MIN

Paperclip: Run Kopi Kaki as a One-Person Company

Build a Human-AI Workforce with Autonomous AI Agents · AI company with an org chart, governed by you · LO3

> You are the only human. A CEO agent hires a Chief of Staff, the Chief of Staff hires the marketing team, and you approve every hire from the Board seat.

The business scenario

You are the founder-CEO of Kopi Kaki Co., a fictional Singapore kopi startup. The Tiong Bahru flagship opens on 1 Dec 2026 with the Gula Melaka Cold Brew (SGD 6.80, #ShiokColdBrew); until then you sell at weekend pop-ups and take app pre-orders. Your team is two baristas — and the AI agents you set up in this course. You have proved that agents can do the work. Now you decide to run Kopi Kaki as a one-person company (OPC): you sit on the Board, and an AI CEO runs the day-to-day. You set up Paperclip, write the CEO's instructions, and let the CEO hire a Chief of Staff, who hires the marketing team. Then you give the team the same launch-video brief as Activity 5 and compare.

You will deliver

- Paperclip running at http://localhost:3100 with the company 'Kopi Kaki Co.'
- CEO agent with written instructions; a Chief of Staff hired through Board approval
- Marketing team hired by the Chief of Staff (researcher, writers, image creator, video producer, reviewer)
- Org chart screenshot: Board → CEO → Chief of Staff → team
- The launch-video task delivered by the team, and comparison.md: Paperclip vs Hermes Kanban

Agents: CEO → Chief of Staff → 6 marketing agents
You (the human): You are the Board: write the mission, approve hires and budgets, accept the video
Goal: Set up a governed AI company with an org chart, hiring approvals and budgets, and compare it with the Hermes Kanban team (LO3).

Before you start

- Node.js 24.11+ (or Docker Desktop)
- One agent CLI signed in on this laptop — Claude Code, Codex, Gemini CLI or the Hermes Agent from Topic 2
- This lab's data/ copied into ~/kopi-kaki

Lab files (data/)

File	What it is
`paperclip-company-setup.md`	Values for the Paperclip company set-up

`brand-guide.md`	Brand voice, colours, platform formats
`campaign-brief.md`	Gula Melaka Cold Brew campaign brief (offer, KPIs, timeline)
`compliance-rules.md`	C1–C10 brand-safety and escalation rules
`video-qa-checklist.md`	V1–V9 video QA checks
`assets/brand-palette.png`	Brand image asset
`assets/cold-brew-product-card.png`	Brand image asset
`assets/kopi-kaki-logo.png`	Brand image asset
`paperclip-opc-setup.md`	Values for the Kopi Kaki one-person company and the target org chart

All company data in this lab (Kopi Kaki Co., people, reviews, comments, metrics, competitors) is fictional and generated for training.

Steps

1. What Paperclip is (5 min)

Paperclip (open source, github.com/paperclipai/paperclip) runs a company of AI agents: an org chart, job titles, reporting lines, budgets, tasks and an approval inbox. You are the Board. Hermes Kanban organised tasks for your team; Paperclip organises the people — who reports to whom, who may hire, and what each agent may spend. Watch the overview first.

TERMINAL

```
# Overview video: https://www.youtube.com/watch?v=WNYe5mD4fY8
node --version          # v24.11.0 or higher
hermes --version        # or: claude --version / codex --version
```

✓ Expected: Node ≥ 24.11 and at least one signed-in agent CLI.

2. Install and start Paperclip (10 min)

Option A (recommended): the one-line onboarder — local, trusted, no login. Option B: Docker Desktop (Windows: enable WSL2 first with `wsl --install`); the compose file needs two secrets, and you will create an account and 'Claim this instance' on first visit. Leave the terminal running. Later restarts: `npx paperclipai run`.

TERMINAL

```
# Option A
npx paperclipai onboard --yes

# Option B (Docker)
git clone https://github.com/paperclipai/paperclip.git && cd paperclip
export BETTER_AUTH_SECRET="$(openssl rand -hex 32)"
export PAPERCLIP_TOOL_ACTION_SIGNING_SECRET="$(openssl rand -hex 32)"
export ANTHROPIC_API_KEY="sk-ant-..."     # and/or OPENAI_API_KEY
docker compose -f docker/docker-compose.quickstart.yml up --build
```

✓ Expected: 'Server running at http://localhost:3100' — open it in your browser.

3. Set up the company: Kopi Kaki Co. (5 min)

In the onboarding wizard name the organization Kopi Kaki Co. Create the first agent and name her Ava. Connect a model (the same CLI you checked). Then set the mission in the organization description and a training budget cap. Leave the onboarding task waiting — hiring rules come first.

TERMINAL

```
npx paperclipai company list                # copy the company id
npx paperclipai budget company:update -C <company-id> --payload-json '{"budgetMonthlyCents":6000}'
```

✓ Expected: Kopi Kaki Co. exists with a USD 60/month cap under Audit → Budgets.

4. Set up the CEO and write her instructions (8 min)

Make Ava the CEO with the CLI, then replace the starter text in Agents → Ava → Instructions → AGENTS.md with the CEO instructions below. Upload brand-guide.md, campaign-brief.md and compliance-rules.md with + (Add agent file). Finally switch ON Settings → General → Hiring → 'Require board approval for new hires'.

TERMINAL

```
npx paperclipai agent list -C <company-id>    # copy Ava's agent id
npx paperclipai agent update <ava-id> --payload-json '{"role":"ceo","title":"CEO"}'
```

PROMPT — PASTE INTO PAPERCLIP → AGENTS → AVA → INSTRUCTIONS → AGENTS.MD

```
You are Ava, CEO of Kopi Kaki Co., a Singapore kopi startup run as a one-person company: the
founder is the Board and the only human.
Mission: launch the Gula Melaka Cold Brew on 1 Dec 2026 - 2,500 app pre-orders, one approved short
video every week, AI spend under USD 60/month, zero compliance breaches.
How you work:
- You do not do the marketing work yourself. You hire and manage.
- Your first hire is a Chief of Staff who runs the marketing team for you.
- Every hire needs a mandate, a reporting line and a monthly budget, and waits for Board approval.
- Every public item passes an independent reviewer and then the Board. Never publish anything
yourself.
- Source of truth: brand-guide.md, campaign-brief.md, compliance-rules.md.
- Report weekly to the Board: progress, spend vs budget, risks, decisions needed.
```

✓ Expected: Ava's role is CEO, her instructions show the mandate and three files, and the hiring gate is ON.

5. CEO task: hire a Chief of Staff (5 min)

Give Ava her first task. She proposes the hire; it lands in your approval inbox.

PROMPT — PASTE INTO PAPERCLIP → NEW TASK (ASSIGN TO AVA)

```
Hire a Chief of Staff for Kopi Kaki Co.
Mandate (3 bullets): turn my goals into weekly plans, hire and manage the marketing team, report
progress and risks to you every Friday.
Reports to: you (CEO). Monthly budget: USD 10. Use the same adapter as you.
Submit the hire for Board approval and tell me when it is waiting.
```

✓ Expected: A 'Hire Agent' card in Pending Approvals.

6. Approve the hire from the Board seat (3 min)

Dashboard → Pending Approvals → View details → See full request. Check mandate, budget and reporting line, then approve. CLI alternative:

TERMINAL

```
npx paperclipai approval list -C <company-id> --status pending
npx paperclipai approval approve <approval-id> --decision-note "Approved: Chief of Staff, USD
10/month"
```

✓ Expected: The Chief of Staff is active and reports to Ava.

7. Chief of Staff task: hire the marketing team (8 min)

Now the Chief of Staff staffs the same team you built by hand in Activities 4 and 5. Approve the hires that fit; send one back with Request revision.

TERMINAL

```
npx paperclipai approval list -C <company-id> --status pending
npx paperclipai approval approve <approval-id> --decision-note "Approved"
npx paperclipai approval request-revision <approval-id> --decision-note "Cap the Video Producer at USD 8/month"
```

PROMPT — PASTE INTO PAPERCLIP → NEW TASK (ASSIGN TO THE CHIEF OF STAFF)

```
Hire the Kopi Kaki marketing team. All report to you.
- Market Researcher (finds facts with sources)
- Social Copywriter (LinkedIn, Facebook and Instagram posts)
- Image Creator
- Video Producer (HyperFrames)
- Voice-over Artist
- Content Reviewer (independent: checks posts and videos against compliance-rules.md and video-qa-checklist.md; can block release)
Give each a 3-bullet mandate and a monthly budget. The whole company must stay under USD 60/month including Ava and you. Submit every hire for Board approval.
```

✓ Expected: Six hire requests; after your decisions the team is active.

8. Show the organization chart (3 min)

Open Agents → Org chart view (icon next to New Agent). Expected: Board (you) → Ava (CEO) → Chief of Staff → six marketing agents. Take a screenshot: this is your one-person company.

9. Get the team to produce the launch video (10 min)

Give the Chief of Staff the same brief as Activity 5 so you can compare like with like. Watch the tasks the Chief of Staff creates and assigns.

PROMPT — PASTE INTO PAPERCLIP → NEW TASK (ASSIGN TO THE CHIEF OF STAFF)

```
Produce the Gula Melaka Cold Brew launch video: 20 seconds, 1080x1920, 4 scenes, narration, facts only from campaign-brief.md and product-catalogue.csv, SGD 6.80, launch 1 Dec 2026, no health or halal claims.
Split it into tasks for the team, with the Content Reviewer checking the final file against video-qa-checklist.md. Ask me to approve the script before rendering and the exact final file before anything is used. Never publish.
```

✓ Expected: Sub-tasks assigned across the team, a script approval request, then a final-file approval request.

10. Compare Paperclip with the Hermes Kanban team (8 min)

Write down what each approach gave you.

PROMPT — PASTE INTO PAPERCLIP → NEW TASK (ASSIGN TO AVA)

```
Write comparison.md for the Board: Paperclip company vs the Hermes Kanban team from Activity 5 for
the same launch video.
Rows: set-up time, who decides the task breakdown, approvals and hiring control, budget control,
visibility (org chart, audit), output quality, my review effort, cost.
End with a recommendation: when should a one-person company use a Kanban team, and when a
Paperclip company?
```

✓ Expected: A table and a clear recommendation.

Verification / checkpoint

- □ http://localhost:3100 shows Kopi Kaki Co.; the hiring gate is ON
- □ Ava is CEO with written instructions
- □ Chief of Staff and six marketing agents hired through Board approvals
- □ Org chart: Board → CEO → Chief of Staff → team
- □ The video task ran with script and final-file approvals; comparison.md written

Checkpoint: The org chart shows Board → CEO → Chief of Staff → marketing team, every hire has a Board approval, and comparison.md answers when to use Kanban and when to use Paperclip.

Troubleshooting

Symptom	Fix
`node` too old	Install Node.js 24.11+ (e.g. from nodejs.org or `nvm install 24`).
Docker: 'BETTER_AUTH_SECRET must be set'	Export both secrets (or put them in docker/.env) before `docker compose up`.
Port 3100 busy	Paperclip falls back to another port — read the URL it prints.
Run test fails	Sign in to the CLI first (e.g. `claude`, `codex login`, `hermes setup`) on the same machine, or paste an API key.
Goals page not in the sidebar	Open /<PREFIX>/goals directly, or Settings → Experimental → Goals Sidebar Link.
Hires joined without approval	The hiring gate was OFF — Settings → General → Hiring. Pause the agent and re-hire.

Stretch challenge

Add a weekly routine (Routines → Create) that asks the Chief of Staff for one new video every Monday at 10:00, still behind your approval. What would you need to trust it to run unattended?